



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and password"
- "Prevent duplicate librarian registration and reject weak passwords"
- "Authenticate librarians against registered credentials"
- "Manage session validation and secure credential handling"
- "Report authentication operations to audit"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection by title and prevent duplicates"
- "Delete registered books"
- "List registered books and consult books by identifier or title"
- "List books available for loan"
- "Report catalog operations to audit"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans and prevent duplicates"
- "Delete registered readers"
- "List registered readers and consult readers by identifier or name"
- "Report reader management operations to audit"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans with loan dates for registered readers"
- "Validate reader eligibility and book availability"
- "Ensure books are loaned to only one reader at a time"
- "Remove loan links when books or readers are deleted"
- "Report loan operations to audit"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structures to reset the environment"
- "Recreate base structure and restore environment to clean state"
- "Coordinate reset operations across functional areas"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two user-provided integers"
- "Calculate Fibonacci sequence values using multiple approaches"
- "Calculate Ackermann function results using multiple approaches"
- "Validate input formats for mathematical operations"
- "Offer service catalog for consulting and testing available endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Receive and record operations from registration"

dependencies:

- "authentication"

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"

- from: "Book Catalog Service"
to: "Loan Management Service"

- from: "Book Catalog Service"
to: "Audit Service"

- from: "Reader Management Service"
to: "Loan Management Service"

- from: "Reader Management Service"
to: "Audit Service"

- from: "Loan Management Service"
to: "Book Catalog Service"

- from: "Loan Management Service"
to: "Reader Management Service"

- from: "Loan Management Service"
to: "Audit Service"

- from: "Environment Management Service"
to: "Authentication Service"

- from: "Environment Management Service"
to: "Book Catalog Service"

- from: "Environment Management Service"
to: "Reader Management Service"

- from: "Environment Management Service"
to: "Loan Management Service"

- from: "Environment Management Service"
to: "Audit Service"

- from: "Audit Service"
to: "authentication"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials with login and password"
- "Create and manage user sessions"
- "Validate active sessions for restricted functionalities"
- "Invalidate sessions on logout"
- "Manage session expiration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search one-way and round-trip flights by origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create bookings for one-way and round-trip flights"
- "Generate unique booking identifiers and record booking dates"
- "Consult and list customer bookings"
- "Cancel existing bookings and update booking status"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Retrieve customer profile contact and address data"
- "Update customer contact and address details"
- "Confirm profile changes using current password"
- "Protect customer personal data against unauthorized access"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present customer loyalty program status"
- "Display accumulated miles for the current year"
- "Display total accumulated miles as read-only information"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload application database with configurable sample customers"
- "Recreate operational data including flights"

dependencies:

- "customers"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Flight Search Service"
to: "destination"
- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "customers"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register cargo with origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and present candidate routes matching cargo criteria"
- "Validate routes against origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Register and validate logistics events for reception"

dependencies:

- "load"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate operational status and tracking situation of cargo"
- "Maintain event history distinguishing expected from unexpected events"
- "Calculate last known location"

dependencies:

- "estimated arrival"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Ingest logistics events through operational interface"

dependencies:

- "integration API"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"

to: "destination"

- from: "Route Planning Service"

to: "destination"

- from: "Logistics Event Registration Service"

to: "load"

- from: "Cargo Tracking Service"

to: "estimated arrival"

- from: "Integration Service"

to: "integration API"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Manage active investor sessions"
- "Process investor logout"
- "Enforce access control for authenticated sessions"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors and validate unique identification"
- "Update investor profile information"

- "Maintain investor account financial balances and access metrics"
- "Process balance debits and credits for trading operations"
- "Provide investor registration and financial data"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes individually and in batch"
- "Provide consolidated market summary"
- "Update traded volume and market indicators after operations"
- "Handle invalid asset codes"

dependencies:

- ""

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create buy and sell trading orders"
- "Apply fixed operation fees"
- "Execute orders immediately or deferred based on configuration"
- "Track order lifecycle status"
- "Prevent duplicate order completion"
- "Cancel sell orders for unavailable positions"
- "Maintain investor order history and recent summaries"
- "Issue alerts for completed orders"

dependencies:

- "Investor Account Service"
- "Portfolio Service"
- "Market Data Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor asset positions"
- "Add new positions from completed buy orders"
- "Remove positions from completed sell orders"
- "Consult portfolio positions and calculate market value"
- "Calculate total invested amount and consolidated gain or loss"
- "Generate consolidated financial position summary"
- "Initiate sales directly from portfolio positions"

dependencies:

- "Trading Service"
- "Market Data Service"
- "Investor Account Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Manage order processing modes and alert preferences"
- "Prepare and rebuild simulation database"
- "Reset operational simulation data"
- "Provide execution statistics and monitoring indicators"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Trading Service"
- "Portfolio Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Trading Service"
to: "Investor Account Service"
- from: "Trading Service"
to: "Portfolio Service"
- from: "Trading Service"
to: "Market Data Service"
- from: "Trading Service"
to: "Simulation Administration Service"
- from: "Portfolio Service"
to: "Trading Service"
- from: "Portfolio Service"
to: "Market Data Service"
- from: "Portfolio Service"
to: "Investor Account Service"
- from: "Simulation Administration Service"
to: "Investor Account Service"
- from: "Simulation Administration Service"
to: "Market Data Service"

- from: "Simulation Administration Service"
to: "Trading Service"
- from: "Simulation Administration Service"
to: "Portfolio Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with username and password"
- "Authenticate registered users"
- "Manage user profiles and access roles"
- "Maintain and terminate user sessions"
- "Prevent duplicate user registrations"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List movies with paginated navigation"
- "Display movie details and cast"
- "Manage movie categories and category-based listings"
- "Register new movies in catalog"
- "Delete movies individually and in batches"
- "Enforce movie title uniqueness"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files up to 2 GB"
- "Store and serve video files for playback"
- "Provide movie file download links"
- "Retrieve video files on demand from local repository"
- "Verify physical video file availability for registered movies"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Manage product categories and catalog navigation"
- "Provide product and item listings"
- "Provide item details and keyword search"
- "Display item stock status"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register customer accounts"
- "Manage customer profile and contact information"
- "Manage customer preferences and favorite categories"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate user credentials"
- "Manage customer sessions and logout"
- "Enforce access control and validate access"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage cart items and update quantities"
- "Consolidate duplicate items in cart"

- "Calculate cart subtotals"
- "Clear cart upon order completion"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Track item stock availability and backorder status"
- "Provide updated stock status consultations"
- "Register stock decrease upon order completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Manage checkout process and order finalization"
- "Process billing delivery and payment information"
- "Generate unique order identifiers and record purchase history"
- "Provide customer order listing and order details consultation"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "Authentication Service"
- from: "Order Service"

to: "Customer Account Service"

- from: "Order Service"

to: "Shopping Cart Service"

- from: "Order Service"

to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owner contact and personal information"
- "Update registered owner details"
- "Search owners by last name prefix or broad query"
- "Display complete profile of selected owner"
- "Provide paginated owner search results"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update registered animal data"
- "Maintain animal information including name"

dependencies:

- "type"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits with date and service description"
- "Suggest current date for new visits"
- "Maintain and display visit history for animals"
- "Coordinate visit assignment with veterinarians"
- "Resume pending internal visit processes after system restart"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "Manage veterinarian profiles and specialties"
- "List veterinarians with pagination and structured data format"
- "Calculate veterinarian workload and inform next expected visit"
- "Automatically assign visits based on lowest future workload"
- "Periodically disregard past visit assignments when calculating workload"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "type"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records including departments"

dependencies:

- "contract types"

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain user personal"

dependencies:

- "contractual"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements"
- "Manage vacation requests and states"
- "Calculate vacation balances"
- "Trigger vacation approval and contract expiration notifications"

dependencies:

- "Identity and Access Service"
- "Administration and Configuration Service"
- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain client-linked projects"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Enforce evidence policies"

dependencies:

- "Workforce Management Service"
- "Identity and Access Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations"

dependencies:

- "contacts"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain financial accounts and entries"
- "Track financial indicators"
- "Import billable hours and costs"

dependencies:

- "Billing Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals"

dependencies:

- "taxes"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and operational areas"

dependencies:

- "File Storage Service"
- "Identity and Access Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"

dependencies:

- "Project and Work Tracking Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate analytical data across business domains"
- "Export business reports"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Maintain notification settings"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files"
- "Manage storage settings"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"
to: "contract types"
- from: "Identity and Access Service"
to: "contractual"
- from: "Workforce Management Service"
to: "Identity and Access Service"
- from: "Workforce Management Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Notification Service"
- from: "Project and Work Tracking Service"
to: "Workforce Management Service"
- from: "Project and Work Tracking Service"
to: "Identity and Access Service"
- from: "Project and Work Tracking Service"
to: "File Storage Service"
- from: "Project and Work Tracking Service"
to: "Notification Service"
- from: "CRM and Proposals Service"
to: "contacts"
- from: "Financial Management Service"
to: "Billing Service"
- from: "Financial Management Service"
to: "Project and Work Tracking Service"
- from: "Billing Service"
to: "taxes"
- from: "Content and Document Management Service"
to: "File Storage Service"
- from: "Content and Document Management Service"
to: "Identity and Access Service"
- from: "Commissioning Service"
to: "Project and Work Tracking Service"
- from: "Commissioning Service"
to: "Workforce Management Service"
- from: "Commissioning Service"

to: "File Storage Service"

- from: "Commissioning Service"

to: "Notification Service"

- from: "Reporting Service"

to: "Administration and Configuration Service"

- from: "Reporting Service"

to: "Identity and Access Service"

- from: "Reporting Service"

to: "Workforce Management Service"

- from: "Reporting Service"

to: "Project and Work Tracking Service"

- from: "Reporting Service"

to: "CRM and Proposals Service"

- from: "Reporting Service"

to: "Financial Management Service"

- from: "Reporting Service"

to: "Billing Service"

- from: "Reporting Service"

to: "Content and Document Management Service"

- from: "Reporting Service"

to: "Commissioning Service"

- from: "Notification Service"

to: "Administration and Configuration Service"

- from: "File Storage Service"

to: "Administration and Configuration Service"